Detailed Bill of Quantities for Juspay Hyperswitch Deployment on Azure (Dev, Staging, Production, DR)

I. Executive Summary

This report outlines a comprehensive Bill of Quantities (BOQ) for deploying Juspay Hyperswitch across development, staging, and production environments, including a robust Disaster Recovery (DR) strategy. The proposed architecture leverages Azure services, strategically deployed across Azure Central India (primary region) and Azure West India (DR region). This design prioritizes enterprise-grade performance, aiming to support 5000 Transactions Per Second (TPS), while ensuring high reliability (targeting 99.999%+ uptime), stringent PCI DSS compliance, and modularity for future extensibility. The selection of services and their configurations incorporates strategies for cost efficiency, such as leveraging autoscaling capabilities, optimizing resource allocation, and considering appropriate service tiers for each environment.

II. Introduction to Juspay Hyperswitch & Core Requirements

Juspay Hyperswitch Overview

Juspay Hyperswitch is an open-source, modular payments platform designed to provide merchants and brands with enhanced control and transparency over their payment infrastructure. Operating under an Apache 2.0 license and holding PCI certification, Hyperswitch offers a fully customizable payment stack, freeing businesses from the limitations often associated with proprietary systems. Its design emphasizes seamless integration with various global payment solutions, enabling merchants to integrate their preferred providers for critical payment functions such as processing, acquiring, fraud management, tokenization, vaulting, and 3DS authentication.

This platform is engineered as the "Linux for Payments," providing full visibility and accelerating integrations. Its modular architecture allows businesses to deploy it either as individual components or as a full-stack orchestration system. Key functionalities include intelligent routing to optimize transaction approvals, customizable checkout experiences, a secure token vault, robust authentication mechanisms, support for alternative payment methods (APMs), cost observability, payouts, reconciliation, and unified analytics. Juspay, the creator of Hyperswitch, is a major player in the payment infrastructure space, processing over 200 million transactions daily with a reliability rate exceeding 99.999% and an annual transaction volume of $670 billion.

Hyperswitch Architecture and Core Dependencies

Hyperswitch's architecture is built around three primary components: the Core, Connectors, and Storage. The Core serves as the central unifying layer, housing the business logic for payment processing, authenticating requests, and executing algorithms to select the optimal payment processor. Connectors are responsible for integrating with various payment processors, handling data transformation, triggering API requests, and interpreting responses, operating as stateless services decoupled from the storage layer. The Storage layer ensures persistence and consistency for transactional data, designed to avoid becoming a bottleneck for high-performance operations. It is further segmented into SQL storage for persistence, an in-memory key-value store (Redis) to offload high-frequency read-write operations from the SQL database, and a drainer to steadily stream SQL queries.

The application services within Hyperswitch consist of the Router and the Scheduler. The Router is the main component, implementing core payment functionalities and managing the coordination of payment processing and routing tasks. The Scheduler handles tasks requiring timed execution, such as automated deletion of saved card details or API key expiry notifications. It comprises a Producer, which tracks tasks and batches them for execution, and a Consumer, which retrieves and executes these batches from a Redis queue.

The foundational technologies underpinning Hyperswitch include:

- **Rust:** The application is predominantly written in Rust (80.8%), chosen for its high performance, low-level control, and memory safety, which are critical for a high-throughput payment system.
- **PostgreSQL:** This relational database is central to Hyperswitch, storing vital data such as customer information, merchant details, and payment-related records. It is typically configured with a master-replica setup to optimize both read and write operations.
- **Redis:** Employed for two main purposes: caching frequently accessed data to minimize application latencies and reduce the load on the primary database, and serving as a queuing mechanism for the Scheduler component.
- **Docker/Podman:** Hyperswitch supports deployment via containerization, utilizing Docker or Podman for environment setup and orchestration.
- **Monitoring:** The "Full" deployment profile of Hyperswitch includes monitoring and schedulers. The system is designed to push metrics and traces in OpenTelemetry Protocol (OTLP) format to an OpenTelemetry collector, with Prometheus used for metrics collection and Promtail/Loki for log aggregation.

Table 1: Juspay Hyperswitch Core Components & Dependencies

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Component Name	Primary Technology	Function/Purpose	Key Features
Core	Rust	Unifying layer, business logic for payment processing, authentication, optimal processor choice.	High performance, reliability, scalability.
Connectors	Rust	Integrations with payment processors, data transformation, API requests/responses.	Stateless, decoupled from storage, standard interface.
Storage	PostgreSQL, Redis	Persistence with consistency for transactional data.	SQL storage for persistence, Redis for high-frequency R/W, drainer for steady SQL writes.
Router	Rust	Main application service, core payment functionalities, managing processing and routing.	Central hub for payment activities, handles requests, processing, routing.
Scheduler	Rust, Redis	Handles scheduled tasks (e.g., card deletion, API key expiry).	Producer (tracks tasks, batches), Consumer (executes tasks from Redis queue).
Vault	Rust, PostgreSQL, Redis	PCI-compliant service to store sensitive payment credentials.	Unified, secure, reusable store of customer-linked payment methods.
Monitoring	OpenTelemetry, Prometheus, Loki, Promtail	Observability for application metrics, traces, and logs.	OTLP format for metrics/traces, Prometheus for metrics, Promtail/Loki for logs.

Performance and Reliability Goals

The user's implicit requirement for a high-throughput payment gateway translates to a target of 5000 Transactions Per Second (TPS). This necessitates a robust and highly performant underlying infrastructure. Juspay's existing operations demonstrate a proven capability to handle over 200 million transactions daily with a reliability rate exceeding 99.999%. The proposed architecture is designed to meet or exceed this standard, ensuring zero downtime and a safe mode for handling exceptions.

Scalability is a paramount concern, with the design supporting horizontal scaling to accommodate peak transaction volumes and handle up to 5K TPS. Furthermore, low latency is critical for payment processing, which is addressed through efficient data retrieval mechanisms, the use of Redis caching, and the high-performance characteristics of Rust as the core development language.

PCI DSS Compliance Mandates

Juspay Hyperswitch is explicitly PCI certified. This certification means that the underlying Azure infrastructure must also inherently support and facilitate PCI DSS compliance. This extends beyond merely securing the application itself to ensuring that every component of the cloud environment—including compute, storage, network, and security services—adheres to the stringent PCI DSS requirements. Azure's own PCI DSS Service Provider Level 1 certification provides a foundational advantage, indicating that the platform itself meets many of these requirements.

However, the responsibility for compliance is shared. While Microsoft ensures the compliance of the underlying cloud infrastructure, the customer remains responsible for how their deployed components are configured and managed to maintain compliance. For instance, if Hyperswitch features a PCI-compliant vault for tokenization , the Azure Key Vault or Managed HSM used to protect the cryptographic keys for that vault must also be PCI compliant. This necessitates careful selection of specific Azure services and tiers, along with their correct configuration, rather than assuming platform-level compliance covers all aspects of the customer's workload. The focus shifts from whether Azure is compliant to how the customer effectively utilizes Azure's compliant services to achieve and maintain compliance for their specific payment processing operations.

III. Azure Deployment Strategy & High-Level Architecture

A. Environment Segregation

To support the full lifecycle of development, testing, and production operations for Juspay Hyperswitch, a clear segregation of environments is essential:

- **Development (Dev):** This environment is primarily focused on cost-effectiveness and enabling rapid iteration for developers. Resources here will be scaled down, potentially utilizing burstable SKUs and shared services where appropriate. This environment is not intended for high-performance testing or high availability, prioritizing flexibility and cost efficiency.
- **Staging (Stag):** The staging environment serves as a near-production replica. Its purpose is to facilitate rigorous testing, including performance benchmarking, integration testing, security audits, and User Acceptance Testing (UAT). The sizing of resources in this environment will closely mirror

production to ensure high fidelity, though it may be slightly scaled down to balance cost with realistic simulation. This environment is critical for validating deployment processes and confirming performance characteristics before a production rollout.

- **Production (Prod):** This is the live environment, designed for mission-critical payment processing. It is engineered for maximum performance, high availability, scalability, and security. All components within the production environment will be deployed with robust redundancy, comprehensive monitoring, and advanced security controls to meet the demands of a high-throughput payment gateway.
- **Disaster Recovery (DR):** A geographically separate region, Azure West India, will serve as the disaster recovery site. This provides business continuity in the event of a major outage in the primary region (Central India). The DR environment will be configured for a warm standby or active-passive model, incorporating continuous data replication and automated failover capabilities to minimize downtime and data loss.

B. Regional Strategy

The selection of Azure Central India as the primary region and Azure West India as the secondary/DR region is a deliberate strategic choice. Both regions are located within India, which is a common requirement for financial technology companies operating in the country dueating to data residency regulations. This ensures that payment data remains within national borders, addressing crucial compliance mandates. Furthermore, opting for geographically proximate regions within the same country offers the advantage of lower inter-region latency compared to cross-continental DR setups. This reduced latency is particularly beneficial for payment systems, where rapid data replication and swift failover capabilities are paramount for maintaining performance and minimizing recovery times during a disaster. This balance between geographic separation for disaster resilience and regional proximity for performance and compliance is a key aspect of the deployment strategy.

C. Architectural Overview (Conceptual Representation)

The proposed architecture visually represents a multi-region deployment designed for resilience and performance.

Primary Region (Azure Central India): This region hosts the active production workload. Key components include:

- **Azure Kubernetes Service (AKS):** Orchestrates the Hyperswitch Router and Control Center, providing a managed environment for containerized applications.

- **Azure Virtual Machines (VMs):** Dedicated VMs are provisioned for the Kafka cluster and Scheduler components, acknowledging their stateful nature and specific performance requirements. Monitoring components are also hosted on VMs.
- **Azure Database for PostgreSQL Flexible Server:** The primary database for persistent storage, configured for high performance and availability.
- **Azure Cache for Redis:** Provides high-speed caching and queuing for the application.
- **Networking:** Azure Application Gateway with Web Application Firewall (WAF) acts as the primary ingress for HTTP/HTTPS traffic. Azure Firewall provides centralized network security and outbound filtering. Internal Load Balancers distribute traffic within the cluster and to backend services.
- **Security Services:** Azure Key Vault Managed HSM for secure key management and Azure DDoS Protection Standard for network-level protection.

Secondary Region (Azure West India): This region serves as the disaster recovery site, configured for warm standby. Key components include:

- **Scaled-down Compute:** AKS node pools and Azure VMs for Kafka and Scheduler are provisioned in a reduced state, ready to scale up rapidly during a failover event.
- **Replicated Database:** Azure Database for PostgreSQL is configured for geo-replication, ensuring continuous data synchronization from the primary region.
- **Replicated Caching:** Azure Cache for Redis is also geo-replicated to maintain cache consistency.
- **Networking & Security:** Similar networking and security services (Application Gateway with WAF, Azure Firewall, DDoS Protection) are deployed in a standby configuration, ready to take over traffic.

Global Services:

- **Azure Front Door Premium:** This critical global service acts as the primary entry point for all user traffic. It intelligently routes requests to the closest healthy backend, ensuring low latency. In a disaster scenario, Azure Front Door automatically detects the unavailability of the primary region and seamlessly fails over to the DR region , ensuring continuous service availability for the payment gateway.

IV. Detailed Bill of Quantities (BOQ) per Environment

This section provides a detailed Bill of Quantities for each environment, outlining specific Azure services, recommended SKUs, quantities, and their justifications. The estimated monthly costs are indicative and would require precise calculation based on current Azure pricing and specific consumption patterns.

A. Development Environment BOQ (Azure Central India)

Purpose: Development, testing, and feature iteration. Optimized for cost efficiency.

Table 2.1: Development Environment BOQ

Azure Service	Recommended SKU/Tier	Key Specifications	Quantity	Justification	Estimated Monthly Cost
Compute					
AKS (Hyperswitch Router/Control Center)	Standard_B2s_v3	2 vCPUs, 8 GiB RAM, 32 GiB Standard SSD	2 nodes	Consistent container orchestration with Prod, cost-effective burstable VMs for dev.	(To be calculated)
Azure Virtual Machine (Kafka)	Standard_D2s_v3	2 vCPUs, 8 GiB RAM, 100 GiB Premium SSD	1 VM	Minimal Kafka instance on VM, aligning with Juspay's preference for stateful workloads.	(To be calculated)
Azure Virtual Machine (Scheduler/Monitoring)	Standard_B2ms	2 vCPUs, 8 GiB RAM, 100 GiB Standard SSD	1 VM	Cost-effective VM for non-critical services.	(To be calculated)
Database					
Azure Database for PostgreSQL Flexible Server	Burstable B2ms	2 vCPUs, 8 GiB RAM, 128 GiB Premium SSD	1 instance	Cost-optimized PostgreSQL	(To be calculated)

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Azure Service	Recommended SKU/Tier	Key Specifications	Quantity	Justification	Estimated Monthly Cost
		v2 (3000 IOPS baseline)		with stop/start capability for dev.	
Caching					
Azure Cache for Redis	Standard C1	1 GB memory, up to 1,000 connections	1 instance	Basic caching for development, non-HA.	(To be calculated)
Networking & Security					
Virtual Network (VNet)	-	Basic network segmentation	1	Foundation for cloud resources.	(Included in other services)
Subnets	-	App, DB, Kafka, Mgmt	4	Logical isolation within VNet.	(Included in other services)
Network Security Groups (NSGs)	-	Basic inbound/outbound rules	4	Basic traffic filtering.	(Included in other services)
Public IP Addresses	Standard	Minimal external access	2	For AKS Load Balancer and jump box.	(To be calculated)
Azure Load Balancer	Standard	For AKS ingress	1	Basic load balancing for AKS.	(To be calculated)
Azure Firewall	Basic	Outbound filtering	1	Basic network security for dev environment.	(To be calculated)

Azure Service	Recommended SKU/Tier	Key Specifications	Quantity	Justification	Estimated Monthly Cost
Azure Key Vault	Standard	Secrets management	1	Secure storage for application secrets.	(To be calculated)
Monitoring & Logging					
Azure Monitor	-	Metrics, logs, alerts	1 instance	Basic monitoring for dev.	(Consumption-based)
Log Analytics Workspace	-	Centralized logging	1 instance	Central log collection.	(Consumption-based)
Total Estimated Monthly Costs (Dev)					**(To be calculated based on specific Azure pricing for chosen SKUs)**

B. Staging Environment BOQ (Azure Central India)

Purpose: Pre-production testing, performance benchmarking, security audits, and UAT. High fidelity to Production.

Table 2.2: Staging Environment BOQ

Azure Service	Recommended SKU/Tier	Key Specifications	Quantity	Justification	Estimated Monthly Cost
Compute					
AKS (Hyperswitch Router/Control Center)	Standard_D4ds_v5	4 vCPUs, 16 GiB RAM, 100 GiB Premium SSD,	3-5 nodes	Realistic load testing, mirroring production scale for	(To be calculated)

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Azure Service	Recommended SKU/Tier	Key Specifications	Quantity	Justification	Estimated Monthly Cost
		Autoscaling (2-5 nodes)		core components.	
Azure Virtual Machine (Kafka Cluster)	Standard_D4ds_v5	4 vCPUs, 16 GiB RAM, 200 GiB Premium SSD	3 VMs	Scaled Kafka cluster for performance testing.	(To be calculated)
Azure Virtual Machine (Scheduler/Monitoring)	Standard_D2ds_v5	2 vCPUs, 8 GiB RAM, 100 GiB Premium SSD	1 VM	Dedicated VM for critical support services.	(To be calculated)
Database					
Azure Database for PostgreSQL Flexible Server	General Purpose D8ds_v5	8 vCPUs, 32 GiB RAM, 256 GiB Premium SSD v2 (68719 IOPS)	1 instance	Scaled for performance testing, zone-redundant HA.	(To be calculated)
Caching					
Azure Cache for Redis	Premium P2	13 GB - 130 GB, up to 15,000 connections, HA (replication)	1 instance	Higher capacity for load testing, HA.	(To be calculated)
Networking & Security					
Virtual Network (VNet)	-	Segregated from Dev, mirroring Prod	1	Production-like network isolation.	(Included in other services)
Subnets	-	App, DB, Kafka, Mgmt	4	Granular segmentation.	(Included in other services)

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Azure Service	Recommended SKU/Tier	Key Specifications	Quantity	Justification	Estimated Monthly Cost
Network Security Groups (NSGs)	-	Granular rules	4	Fine-grained traffic control.	(Included in other services)
Azure Application Gateway v2 with WAF_v2	WAF_v2	Autoscaling (min 2, max 10 instances)	1 instance	L7 load balancing, WAF protection for security audits and UAT.	(To be calculated)
Azure Firewall	Standard	Robust network security, outbound filtering	1 instance	Enhanced security for pre-production.	(To be calculated)
Azure Key Vault	Premium	Secure secrets management, mirroring Prod's security posture	1 instance	PCI-compliant key storage for testing.	(To be calculated)
Azure DDoS Protection	Standard (Network Protection)	VNet protection during load testing	1 plan	Protection during security assessments and performance tests.	(To be calculated)
Monitoring & Logging					
Azure Monitor	-	Comprehensive monitoring	1 instance	Detailed observability for testing.	(Consumption-based)

Azure Service	Recommended SKU/Tier	Key Specifications	Quantity	Justification	Estimated Monthly Cost
Log Analytics Workspace	-	Centralized logging	1 instance	Central log collection for analysis.	(Consumption-based)
Total Estimated Monthly Costs (Staging)					**(To be calculated based on specific Azure pricing for chosen SKUs)**

C. Production Environment BOQ (Azure Central India)

Purpose: Mission-critical payment processing. Maximum performance, high availability, scalability, and security.

Table 2.3: Production Environment BOQ (Central India)

Azure Service	Recommended SKU/Tier	Key Specifications	Quantity	Justification	Estimated Monthly Cost
Compute					
AKS System Node Pool	Standard_D8ds_v5	8 vCPUs, 32 GiB RAM, 100 GiB Premium SSD	3 nodes	Dedicated for Kubernetes system components, ensuring stability.	(To be calculated)
AKS User Node Pool (Router)	Standard_D16ds_v5	16 vCPUs, 64 GiB RAM, 200 GiB Premium SSD, Autoscaling (6-10 nodes)	6-10 nodes	Sized for 5000 TPS (approx. 84 vCPUs needed), with buffer and autoscaling	(To be calculated)

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Azure Service	Recommended SKU/Tier	Key Specifications	Quantity	Justification	Estimated Monthly Cost
				for peak loads.	
Azure Virtual Machine (Kafka Cluster)	Standard_E8ds_v5	8 vCPUs, 64 GiB RAM, 500 GiB Premium SSD	3 VMs	High-memory VMs for stateful Kafka, aligning with Juspay's experience for performance and cost efficiency.	(To be calculated)
Azure Virtual Machine (Scheduler)	Standard_D4ds_v5	4 vCPUs, 16 GiB RAM, 100 GiB Premium SSD	2 VMs	Active-active for HA of critical scheduler components.	(To be calculated)
Azure Virtual Machine (Monitoring)	Standard_D2ds_v5	2 vCPUs, 8 GiB RAM, 100 GiB Premium SSD	2 VMs	For robust monitoring stack (Prometheus, Loki, OpenTelemetry Collector).	(To be calculated)
Database					
Azure Database for PostgreSQL Flexible Server (Primary)	Memory Optimized E32ds_v5	32 vCPUs, 256 GiB RAM, 512 GiB Premium SSD v2 (80,000 IOPS, 1200 MB/s throughput), Zone-	1 instance	High-performance tier for critical, high-frequency R/W payment data, ensuring	(To be calculated)

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Azure Service	Recommended SKU/Tier	Key Specifications	Quantity	Justification	Estimated Monthly Cost
		Redundant HA		consistency and transactional integrity.	
Azure Database for PostgreSQL Flexible Server (Read Replicas)	Memory Optimized E16ds_v5	16 vCPUs, 128 GiB RAM, 256 GiB Premium SSD v2 (80,000 IOPS)	2 instances	Offload read traffic, enhance performance and availability.	(To be calculated)
Caching					
Azure Cache for Redis	Premium P3	26 GB - 260 GB, up to 30,000 connections, Redis Cluster, Persistence, Multi-replica HA	1 instance	High-speed data retrieval, session management, and queuing, critical for 5000 TPS, with HA and data durability.	(To be calculated)
Networking & Security					
Virtual Network (VNet)	-	Robust segmentation	1	Core network isolation for production.	(Included in other services)
Subnets	-	Dedicated for App, DB, Kafka, Mgmt, Private Endpoints	5	Strict network segmentation for security.	(Included in other services)
Network Security Groups (NSGs)	-	Granular rules	5	Fine-grained traffic control	(Included in other services)

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Azure Service	Recommended SKU/Tier	Key Specifications	Quantity	Justification	Estimated Monthly Cost
				between tiers.	
Azure Private Link	-	Secure private connectivity for PaaS services	1	Bypasses public internet for PostgreSQL, Redis, Key Vault.	(To be calculated)
Azure Application Gateway v2 with WAF_v2	WAF_v2	Autoscaling (min 5, max 125 instances)	1 instance	Primary ingress for L7 traffic, WAF for L7 attack protection, critical for 5000 TPS.	(To be calculated)
Azure Firewall	Premium	Centralized network security, IDPS (Deny mode), TLS inspection, 10 Gbps throughput	1 instance	Advanced threat protection, enforcing network policies, essential for PCI DSS.	(To be calculated)
Azure Load Balancers	Standard	Internal load balancing for AKS & Kafka/Scheduler VMs	2 instances	Distributes traffic within the cluster and to backend services.	(To be calculated)
Azure Front Door	Premium	Global traffic management, WAF, SSL offloading, caching, failover to DR	1 instance	Global entry point, low-latency routing, and seamless multi-region failover.	(To be calculated)

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Azure Service	Recommended SKU/Tier	Key Specifications	Quantity	Justification	Estimated Monthly Cost
Azure Key Vault Managed HSM	Managed HSM	FIPS 140-2 Level 3, PCI compliant, single-tenant HSM	1 instance	Essential for storing sensitive cryptographic keys for Hyperswitch's PCI-compliant vault.	(To be calculated)
Azure DDoS Protection	Standard (Network Protection Plan)	VNet and public IP protection	1 plan	Protects against volumetric, protocol, and application layer DDoS attacks.	(To be calculated)
Azure Policy	-	Enforce organizational standards and compliance	1 subscription	Ensures continuous compliance with security and regulatory mandates.	(Consumption-based)
Azure Security Center/Defender for Cloud	Standard	Continuous security posture management, threat protection	1 subscription	Comprehensive security monitoring and vulnerability management.	(Consumption-based)
Monitoring & Logging					
Azure Monitor	-	Metrics, logs, traces, Application Insights	1 instance	Comprehensive observability for all services and	(Consumption-based)

Azure Service	Recommended SKU/Tier	Key Specifications	Quantity	Justification	Estimated Monthly Cost
				Hyperswitch components.	
Log Analytics Workspace	-	Centralized logging for all application and infrastructure logs	1 instance	Central repository for audit and operational logs.	(Consumption-based)
Total Estimated Monthly Costs (Production - Central India)					**(To be calculated based on specific Azure pricing for chosen SKUs)**

D. Disaster Recovery (DR) Environment BOQ (Azure West India)

Purpose: Business continuity and disaster recovery. Configured for warm standby / active-passive.

Table 2.4: Disaster Recovery Environment BOQ (West India)

Azure Service	Recommended SKU/Tier	Key Specifications	Quantity	Justification	Estimated Monthly Cost
Compute					
AKS System Node Pool	Standard_D8ds_v5	8 vCPUs, 32 GiB RAM, 100 GiB Premium SSD	3 nodes	Standby for Kubernetes system components, ready to scale up.	(To be calculated)
AKS User Node Pool (Router)	Standard_D8ds_v5	8 vCPUs, 32 GiB RAM,	3-5 nodes	Scaled-down for	(To be calculated)

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Azure Service	Recommended SKU/Tier	Key Specifications	Quantity	Justification	Estimated Monthly Cost
		100 GiB Premium SSD, Autoscaling (3-5 nodes)		warm standby, ready to handle traffic during failover.	
Azure Virtual Machine (Kafka Cluster)	Standard_E4ds_v5	4 vCPUs, 32 GiB RAM, 250 GiB Premium SSD	3 VMs	Scaled-down Kafka, ready to scale up for DR operations.	(To be calculated)
Azure Virtual Machine (Scheduler/Monitoring)	Standard_D2ds_v5	2 vCPUs, 8 GiB RAM, 100 GiB Premium SSD	1 VM	Standby for critical support services.	(To be calculated)
Database					
Azure Database for PostgreSQL Flexible Server	Memory Optimized E32ds_v5	32 vCPUs, 256 GiB RAM, 512 GiB Premium SSD v2 (80,000 IOPS, 1200 MB/s throughput)	1 instance	Geo-replicated read replica, capable of promotion to primary during failover. This ensures a low Recovery Point Objective (RPO) for critical payment data, as a 12-hour RPO from	(To be calculated)

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Azure Service	Recommended SKU/Tier	Key Specifications	Quantity	Justification	Estimated Monthly Cost
				cross-region restore would be insufficient for a payment gateway.	
Caching					
Azure Cache for Redis	Premium P3	26 GB - 260 GB, geo-replication, Multi-replica HA	1 instance	Geo-replicated instance for cache consistency and availability in DR.	(To be calculated)
Networking & Security					
Virtual Network (VNet)	-	Mirror primary region's segmentation	1	Network foundation for DR.	(Included in other services)
Subnets	-	Dedicated for App, DB, Kafka, Mgmt, Private Endpoints	5	Strict network segmentation.	(Included in other services)
Network Security Groups (NSGs)	-	Granular rules	5	Fine-grained traffic control.	(Included in other services)
Azure Private Link	-	Secure private connectivit	1	Bypasses public internet for PostgreSQL,	(To be calculated)

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Azure Service	Recommended SKU/Tier	Key Specifications	Quantity	Justification	Estimated Monthly Cost
		y for PaaS services		Redis, Key Vault.	
Azure Application Gateway v2 with WAF_v2	WAF_v2	Autoscaling (min 2, max 10 instances)	1 instance	Standby ingress for L7 traffic, ready to scale up.	(To be calculated)
Azure Firewall	Premium	Standby network security, IDPS (Deny mode), TLS inspection	1 instance	Advanced threat protection for DR environment.	(To be calculated)
Azure Load Balancers	Standard	Internal load balancing for standby components	2 instances	Distributes traffic within the cluster and to backend services.	(To be calculated)
Azure Key Vault Managed HSM	Managed HSM	Replicated or synchronized for key availability in DR	1 instance	Ensures cryptographic keys are available during DR scenario.	(To be calculated)
Azure DDoS Protection	Standard (Network Protection Plan)	Covers DR VNet and public IPs	1 plan	Protects DR environment during failover.	(To be calculated)
Monitoring & Logging					
Azure Monitor	-	Metrics, logs, traces	1 instance	Monitoring for DR environment.	(Consumption-based)
Log Analytics Workspace	-	Centralized logging	1 instance	Central log collection for DR.	(Consumption-based)

Azure Service	Recommended SKU/Tier	Key Specifications	Quantity	Justification	Estimated Monthly Cost
Total Estimated Monthly Costs (DR - West India)					**(To be calculated based on specific Azure pricing for chosen SKUs)**

V. Sizing and Performance Justification

The proposed Azure architecture is meticulously designed to meet the demanding performance and reliability requirements of a payment gateway, specifically targeting 5000 Transactions Per Second (TPS). The selection of each service and its SKU is directly tied to these objectives, ensuring optimal resource utilization and robust operational capabilities.

Detailed Rationale for Chosen SKUs

- **Compute (AKS & VMs):**
  - **AKS for Hyperswitch Router:** The Hyperswitch Router is capable of processing up to 60 payments per second on a single-core processor. To achieve a target of 5000 TPS, approximately 83.33 vCPUs are theoretically required (5000 TPS / 60 TPS per vCPU). For the Production environment, the recommendation of 6 to 10

    Standard_D16ds_v5 nodes for the user node pool (each with 16 vCPUs) provides a total of 96 vCPUs at minimum. This over-provisioning offers a crucial buffer for operational overhead, potential transaction complexity variations, and future growth. The Standard_D16ds_v5 SKU is selected for its balance of CPU, memory, and local SSD, which is beneficial for temporary storage and caching, contributing to overall performance. Furthermore, implementing robust autoscaling for AKS node pools is critical. This ensures the platform can dynamically adapt to unexpected traffic spikes without manual intervention, maintaining high availability and consistent performance under fluctuating loads. The AKS Standard tier for the control plane is chosen to ensure a Service Level Agreement (SLA) for production-critical workloads.

- **Azure Virtual Machines for Kafka:** Juspay's own experience of migrating Kafka from Kubernetes to Amazon EC2 instances due to challenges with stateful workloads, rising costs, and operational complexity strongly informs the decision to deploy Kafka on dedicated Azure Virtual Machines. This approach aligns with their proven strategy for better resource allocation, simplified operations, and cost savings for their core eventing layer. The

  Standard_E8ds_v5 VMs, with their high memory-to-core ratio , are well-suited for data-intensive workloads like Kafka, ensuring ample memory for efficient message processing and storage. This choice acknowledges that while managed Kubernetes (AKS) excels for stateless applications, a more granular control offered by VMs can yield superior results for specific stateful components in a complex system.

- **Database (Azure Database for PostgreSQL Flexible Server):**
  - The Memory Optimized E32ds_v5 SKU for the primary database, paired with 512 GiB of Premium SSD v2 storage and configured for 80,000 IOPS and 1200 MB/s throughput , is selected to meet the critical data persistence and consistency requirements of a payment gateway. Hyperswitch's architecture includes an "in-memory key-value store to debottleneck the SQL storage" and a "drainer to stream and execute SQL read-write queries onto the SQL storage in a steady manner". This design indicates that while PostgreSQL is the ultimate persistent layer, high-frequency operations are offloaded to Redis. Nevertheless, the PostgreSQL database must handle the aggregated load from the drainer and maintain transactional integrity under peak conditions. The Premium SSD v2 is crucial for achieving the low latency and high IOPS necessary for timely transaction commits and data retrieval in a payment system. The Flexible Server's features, including zone-redundant high availability and scaling capabilities , further enhance the database's reliability and performance.
- **Caching (Azure Cache for Redis):**
  - The Premium P3 tier for Azure Cache for Redis is chosen for its ability to handle a high volume of client connections (up to 30,000) and its superior performance, which is essential for high-speed data retrieval, session management, and queuing operations critical for a 5000 TPS payment gateway. This tier also offers crucial features such as Redis Cluster for sharding data across multiple nodes to achieve higher performance and capacity, and Redis persistence (RDB backup) for data durability. Multi-replica support within the Premium tier ensures high availability and resilience for the caching layer.

- **Networking (Application Gateway WAF_v2, Azure Firewall Premium, Azure Front Door):**
  - **Azure Application Gateway WAF_v2:** This service is critical as the primary ingress for HTTP/HTTPS traffic. Its autoscaling capability, allowing it to scale up to 125 instances , and its improved performance for handling requests are vital for managing the high L7 traffic volume associated with 5000 TPS. The integrated WAF provides essential protection against common web vulnerabilities.
  - **Azure Firewall Premium:** Selected for its advanced threat protection capabilities, including Intrusion Detection and Prevention System (IDPS) operating in Deny mode and TLS inspection for outbound traffic. This tier provides a high throughput of up to 10 Gbps even with TLS and IDPS enabled , ensuring secure and high-performance network segmentation. Its features are directly mandated by the stringent security requirements of a PCI DSS compliant environment.
  - **Azure Front Door Premium:** This global service is indispensable for managing traffic across multiple regions. It provides low-latency routing by directing users to the closest available endpoint, performs SSL offloading, and offers caching capabilities. Most importantly, it enables seamless multi-region failover between Central India (primary) and Azure West India (DR), ensuring continuous service availability during regional outages.

Discussion on Autoscaling, Replication, and Performance Optimization Techniques

The architecture incorporates several key strategies to ensure optimal performance, scalability, and reliability:

- **Autoscaling:** Dynamic scaling is fundamental to handling fluctuating transaction volumes. AKS node pool autoscaling allows the compute resources for the Hyperswitch Router to expand or contract based on demand, preventing performance bottlenecks during peak loads and optimizing costs during off-peak hours. Similarly, Azure Application Gateway's autoscaling ensures that the ingress layer can handle sudden surges in L7 traffic. Azure Firewall also gradually scales out when its throughput or CPU utilization reaches a certain threshold , providing adaptive network security capacity.
- **Replication:** High availability and data durability are achieved through robust replication mechanisms. Azure Database for PostgreSQL Flexible Server is configured with master-replica setups within the primary region and geo-replication to the DR region. This ensures data redundancy and enables rapid failover with minimal data loss. Azure Cache for Redis also utilizes multi-replica support to maintain high availability for the caching layer.

- **Performance Optimization:** The choice of Rust as the primary language for Hyperswitch inherently contributes to high performance due to its efficiency and memory safety. The strategic use of Redis for caching significantly reduces database load and application latencies by serving frequently accessed data from memory. The deployment of PostgreSQL with Premium SSD v2 storage ensures high IOPS and throughput for persistent data operations. Furthermore, leveraging Azure's global network edge services like Azure Front Door minimizes latency for end-users by routing requests to the closest point of presence.

Table 3: Sizing and Performance Metrics for Production Environment (5000 TPS)

Component	Target Metric	Azure SKU/Configuration	Achieved Metric/Capacity	Reference
Hyperswitch Router (Compute)	5000 TPS	AKS User Node Pool: 6-10 x Standard_D16ds_v5 nodes	96-160 vCPUs (theoretical max)	
PostgreSQL Database	High-frequency R/W, Consistency	Memory Optimized E32ds_v5 with 512 GiB Premium SSD v2 storage	80,000 IOPS, 1200 MB/s throughput	
Azure Cache for Redis	High-speed caching, 5000 TPS support	Premium P3 (26 GB - 260 GB)	Up to 30,000 connections	
Application Gateway WAF	L7 Traffic Handling	WAF_v2, Autoscaling (min 5, max 125 instances)	Up to 8x more RPS, 16x larger request sizes (compared to previous engine)	
Azure Firewall	Network Throughput	Premium SKU with TLS/IDPS enabled	10 Gbps (with IDPS in Deny mode)	

VI. Security and Compliance Deep Dive

Meeting PCI DSS compliance requirements is paramount for a payment gateway like Juspay Hyperswitch. The proposed Azure architecture integrates comprehensive security measures and utilizes specific Azure services to ensure adherence to these stringent standards.

PCI DSS Implementation Details Across Azure Services

- **Shared Responsibility Model:** Azure maintains PCI DSS Service Provider Level 1 validation , meaning the underlying cloud infrastructure meets many of the compliance requirements. However, the responsibility for securing components deployed

  on Azure is shared. The customer is accountable for configuring and managing their applications and services to maintain compliance.

- **Data Protection:**
  - **Tokenization/Vaulting:** Hyperswitch includes a PCI-compliant vault service for storing sensitive card data. To protect the cryptographic keys used by this vault, Azure Key Vault Managed HSM is deployed. This service provides FIPS 140-2 Level 3 validation and is PCI DSS compliant , offering a single-tenant hardware security module (HSM) for enhanced key protection.
  - **Encryption at Rest:** All data stored within the environment will be encrypted at rest. This includes data in Azure Database for PostgreSQL (utilizing Azure managed disks with encryption), Azure Cache for Redis, and any associated storage accounts. Azure's native encryption capabilities will be leveraged, with options for customer-managed keys via Key Vault for greater control.
  - **Encryption in Transit:** All communication, both internal (between application components, to databases, and to caching layers) and external (to payment processors), will enforce TLS/SSL. Azure Application Gateway and Azure Firewall Premium are configured to support TLS offloading and inspection for secure data transmission.

Network Segmentation and Isolation

Robust network segmentation is a cornerstone of PCI DSS compliance, limiting the scope of a breach.

- **Virtual Network (VNet) & Subnets:** The architecture implements strict network segmentation using Azure VNets and dedicated subnets for each logical tier (e.g., application, database, Kafka, management, private endpoints). This approach isolates components and restricts lateral movement within the network in the event of a security incident.
- **Network Security Groups (NSGs):** Granular NSG rules are applied to control traffic flow between these subnets, permitting only the necessary ports and protocols required for legitimate communication. This "least privilege" network access model enhances security.
- **Azure Firewall Premium:** A centralized Azure Firewall Premium instance acts as the primary network security control for all inbound and outbound traffic. It enforces network policies, provides Layer 3–Layer 7 filtering, and includes

advanced threat protection features like IDPS operating in Deny mode. This ensures that malicious traffic is blocked before it can reach internal systems.

- **Azure Private Link:** To further enhance security, all communication with Azure Platform-as-a-Service (PaaS) services, such as Azure Database for PostgreSQL, Azure Cache for Redis, and Azure Key Vault, will occur over Azure Private Link. This establishes private endpoints within the VNet, ensuring that traffic to these services traverses the secure Azure backbone network, eliminating exposure to the public internet.

Threat Protection and DDoS Mitigation

- **Azure DDoS Protection Standard (Network Protection Plan):** Given the nature of a payment gateway, it is a prime target for Distributed Denial of Service (DDoS) attacks. The Azure DDoS Protection Standard plan (Network Protection tier) is essential for safeguarding the VNet and all public IP resources from volumetric, protocol, and application layer DDoS attacks.
- **Azure Application Gateway WAF_v2:** Beyond network-level protection, the Application Gateway with WAF_v2 provides Layer 7 protection against common web vulnerabilities (e.g., OWASP Top 10) and application-layer attacks.
- **Azure Security Center/Defender for Cloud:** These services provide continuous threat detection, vulnerability management, and security posture assessment across the entire Azure environment, offering a unified view of security status and actionable recommendations.

Identity and Access Management (IAM)

- **Azure Active Directory (Azure AD):** Centralized identity management for all users and services ensures consistent authentication and authorization policies.
- **Role-Based Access Control (RBAC):** A strict least-privilege access model is implemented using Azure RBAC for all Azure resources, ensuring that users and services only have the permissions necessary to perform their designated functions.
- **Managed Identities:** Azure Managed Identities are utilized for Azure services to authenticate to other Azure services securely, eliminating the need for hardcoded credentials and reducing the risk of credential compromise.

Table 4: Security and Compliance Feature Mapping (PCI DSS)

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BOQ for Hyper Switch

PCI DSS Requirement	Azure Service/Feature	Specific SKU/Configuration	Justification/Compliance Benefit	Reference
Data Encryption (at Rest)	Azure Managed Disks (PostgreSQL), Azure Cache for Redis, Storage Accounts	Native encryption, Customer-Managed Keys via Key Vault	Ensures sensitive data is encrypted when stored, meeting PCI DSS requirement 3.4.	
Data Encryption (in Transit)	Azure Application Gateway, Azure Firewall Premium	TLS/SSL enforcement, TLS inspection	Encrypts all data in transit, protecting cardholder data across networks, meeting PCI DSS requirement 4.1.	
Key Management	Azure Key Vault Managed HSM	Managed HSM (FIPS 140-2 Level 3, PCI compliant)	Securely stores and manages cryptographic keys for tokenization/vaulting, meeting PCI DSS requirement 3.5, 3.6.	
Network Segmentation	Virtual Networks (VNets), Subnets, Network Security Groups (NSGs)	Dedicated subnets for tiers, granular NSG rules	Isolates cardholder data environment (CDE) and limits network access, meeting PCI DSS requirement 1.1, 1.2.	(General Azure best practice)
Advanced Network Security	Azure Firewall Premium	IDPS in Deny mode, L3-L7 filtering	Provides advanced threat protection and enforces strict network policies, meeting PCI DSS requirement 1.3, 2.2, 6.4.	
Secure Connectivity	Azure Private Link	Private Endpoints for PaaS services	Ensures private, secure communication to Azure PaaS services, minimizing public exposure, meeting PCI	

PCI DSS Requirement	Azure Service/Feature	Specific SKU/Configuration	Justification/Compliance Benefit	Reference
			DSS requirement 2.2, 4.1.	
DDoS Protection	Azure DDoS Protection Standard (Network Protection Plan)	VNet and public IP protection	Protects public-facing assets from volumetric attacks, ensuring availability, meeting PCI DSS requirement 2.2, 6.4.	
Web Application Firewall (WAF)	Azure Application Gateway WAF_v2	WAF_v2	Protects web applications from common attacks (OWASP Top 10), meeting PCI DSS requirement 6.4.	
Centralized Logging & Monitoring	Azure Monitor, Log Analytics Workspace	Comprehensive logging for all services	Facilitates security event logging, monitoring, and incident response, meeting PCI DSS requirement 10.	
Identity & Access Management	Azure Active Directory, RBAC, Managed Identities	Least-privilege access, strong authentication	Controls access to CDE components and sensitive data, meeting PCI DSS requirement 7, 8.	(General Azure best practice)

VII. Recommendations and Operational Considerations

Beyond the initial Bill of Quantities, successful deployment and long-term operation of Juspay Hyperswitch on Azure require careful consideration of ongoing management, automation, and optimization strategies.

Monitoring, Logging, and Alerting Strategy

A robust observability framework is critical for a high-throughput payment system. The architecture will leverage Azure Monitor, Log Analytics Workspace, and Application Insights to provide comprehensive collection of metrics, logs, and traces across all Azure services and Hyperswitch components. Hyperswitch's native OpenTelemetry integration will be fully utilized to push application-specific metrics and traces, enabling deep insights into payment flows and system health. Custom dashboards will be created for real-time visibility into key performance indicators (KPIs), operational health, and security posture. Proactive alerts will be configured for critical thresholds, anomalies, and potential security incidents, ensuring rapid response to any issues.

Deployment Automation and CI/CD

To ensure consistency, repeatability, and efficiency across all environments, infrastructure as Code (IaC) will be implemented using tools like Azure Resource Manager (ARM) templates, Bicep, or Terraform. This approach allows for version control of infrastructure configurations and automated provisioning. Robust Continuous Integration/Continuous Delivery (CI/CD) pipelines, utilizing platforms such as Azure DevOps or GitHub Actions, will automate the build, testing, and deployment processes for both Hyperswitch application components and infrastructure changes. This reduces manual errors, accelerates release cycles, and maintains environmental consistency.

Ongoing Management and Cost Optimization

Managing cloud costs effectively is an ongoing process. For predictable, long-running workloads in Production and DR environments, Azure Reserved Instances or Savings Plans should be considered. These commitment-based pricing models can significantly reduce compute costs compared to pay-as-you-go rates. Continuous monitoring of resource utilization is essential to identify and right-size VMs, AKS node pools, and database/cache tiers, preventing over-provisioning and unnecessary expenditure. Furthermore, implementing automated shutdown schedules for Development and Staging environments during non-working hours can yield substantial cost savings. Azure Cost Management + Billing tools will be utilized for detailed cost analysis, budgeting, and anomaly detection, providing visibility and control over cloud spending.

It is important to acknowledge that while fully managed services often promise ease of use and cost efficiency, this is not universally true for all components in a complex system. Juspay's experience of moving Kafka from Kubernetes to dedicated Virtual Machines (EC2 in their case, Azure VMs in this proposal) demonstrates that for highly stateful, performance-critical components like Kafka, a more granular control offered by VMs, combined with custom automation (like Juspay's in-house Kafka Controller), can lead to superior operational efficiency and cost savings. This underscores that the optimal choice for each component in a FinTech solution may involve a hybrid approach, rather than a blanket application of a single technology, requiring careful evaluation of workload needs versus service management overhead.

Backup and Restore Strategy

A comprehensive backup and restore strategy is vital for data protection and disaster recovery. For Azure Database for PostgreSQL, automated backups with appropriate retention policies will be configured. For disaster recovery, leveraging geo-replication for PostgreSQL is recommended to achieve a low Recovery Point Objective (RPO) and Recovery Time Objective (RTO). This is critical for a payment gateway, where minimal data loss and rapid recovery are non-negotiable. For Azure Cache for Redis, persistence (RDB backup) will be enabled for Premium tier caches to ensure data durability. Finally, Azure Backup will be implemented for Kafka and Scheduler Virtual Machines to protect these critical stateful components.